

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

IV Semester of Bachelor in Medical Imaging Technology May 2018

BMRIT401 Physics of Newer Imaging Modalities

Date: 21/05/2018 Day: Monday Time: 10.00 am to 10.30 am Maximum Marks: 20

MCQ

Student's ID No.	Answer Sheet for Multiple Choice Questions (MCQs), True or false and Match the following										
<table border="1"><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table>											Examination : BMIT/BOTAT/BOP
Student's Name:	Semester: IV										
Student's Sign :	Section :										
Supervisor's Name :	Course Code : _____										
Supervisor's Sign:	Course Name: _____										

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q – I (A)	Choose the correct answer for the following questions.	10
1.	Longitudinal magnetization also known as	
	a. T1 Relaxation	b. T2 Relaxation
	c. T2* Decay	d. Free induction Decay
2.	Unit of magnetic field is	
	a. RAD	b. Sievert
	c. Tesla	d. Curie
3.	An ultrasound control which eliminates the smaller amplitude voltage pulses produced by weaker echo is known as	
	a. Rejection	b. Amplification
	c. Compensation	d. Demodulation
4.	Niche magnets are designed for	
	a. Muscular Imaging	b. Venous imaging
	c. Arterial Imaging	d. Orthopaedic Imaging

5.	Surface coils can be used for		
	a. Knee imaging	b. Shoulder Imaging	
	c. Abdomen Imaging	d. Ankle Imaging	
6.	Matrix array detectors also referred as		
	a. Hybrid detectors	b. Adaptive detectors	
	c. Fixed array detectors	d. Dual row detectors	
7.	The average speed of ultrasound in body is		
	a. 1510m/s	b. 1520m/s	
	c. 1530m/s	d. 1540m/s	
8.	The principle on which computed Radiography works call		
	a. Photographic luminescence	b. Phosphor luminescence	
	c. Photographic Fluorescence	d. Phosphor fluorescence	
9.	Protons are positively charged and have rotatory movement calls		
	a. Spin	b. Dipole	
	c. Pole	d. Antipole	
10.	DICOM stands for		
	a. Demand in Imaging and Communication for Medicine	b. Digital Imaging and Communications in Medicine	
	c. Digital Imaging in Cancer and Medicine	d. Data Invert and Collection in Medicine	

Q - I (B)	Match the following column “A” with Column “B”.		5
Column A	Answer	Column B	
1. CT reconstruction Technique		a. Photon Starvation	
2. Anti-scatter grid		b. Mammography	
3. Computed Radiography		c. Road Mapping	
4. Digital Subtraction Angiography		d. Fourier transformation	
5. CT artifact		e. Indirect Digital Radiography Technique	

Q - I (C)	Select whether following statement is true or false.	5
		True/False
1.	CT gantry can be tilt to avoid radiation exposure to the patient.	
2.	HIS, RIS, PACS and Tele radiology basically reduce patient discomfort.	
3.	Other than hydrogen, carbon, Sodium, Nitrogen, oxygen molecules cannot be used to generate MRI images.	
4.	Ultrasound waves travel in a straight line.	
5.	Mammography gives high radiation dose to the body as compared to CT.	

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Instructions:

1. Section I and II must be attempted in SINGLE ANSWER SHEET.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION –I		
Q - II Answer the following questions as directed. (Any ten)		20
1.	Enlist CT detectors.	
2.	Define Resonance.	
3.	Name Supplementary views for mammography.	
4.	Define precessional frequency.	
5.	Define Piezoelectric effect.	
6.	Write two applications of Tele Radiology.	
7.	Write two characteristics of Ultrasound beam.	
8.	Define curie temperature.	
9.	Define PD weighting.	
10.	Define T and Z scores.	
11.	Applications of DSA.	
SECTION - II		
Q-III Answer the following questions as directed (Any four)		20
1.	Draw a labeled diagram of USG Transducer.	
2.	Explain any five CT artifacts.	
3.	Describe the basic principle of MRI.	
4.	Write difference between Continuous and pulsed wave Doppler.	
5.	Explain instrumentation of Mammography.	
Q-IV Answer the following questions as directed		10
1.	What is transducer? Explain in Detail with diagram types of ultrasound transducers.	